Presentation

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Recent work：

1. *IDE RankNet*
2. *R2CNN*
3. *PixelLink*

IDE RankNet

Figure 1. Dataset created by Radu et al. Each image has a corresponding difficulty score which is ranged between two and nine. The harder it is to identify the object within the image, the higher the corresponding difficulty score, and vice versa.

Ionescu, Radu Tudor, et al. "How Hard Can It Be? Estimating the Difficulty of Visual Search in an Image." computer vision and pattern recognition (2016): 2157-2166.

IDE RankNet

1. MTL OR MOL

2. RE-Sampling

3.Application: Active Learning or Learning to teach

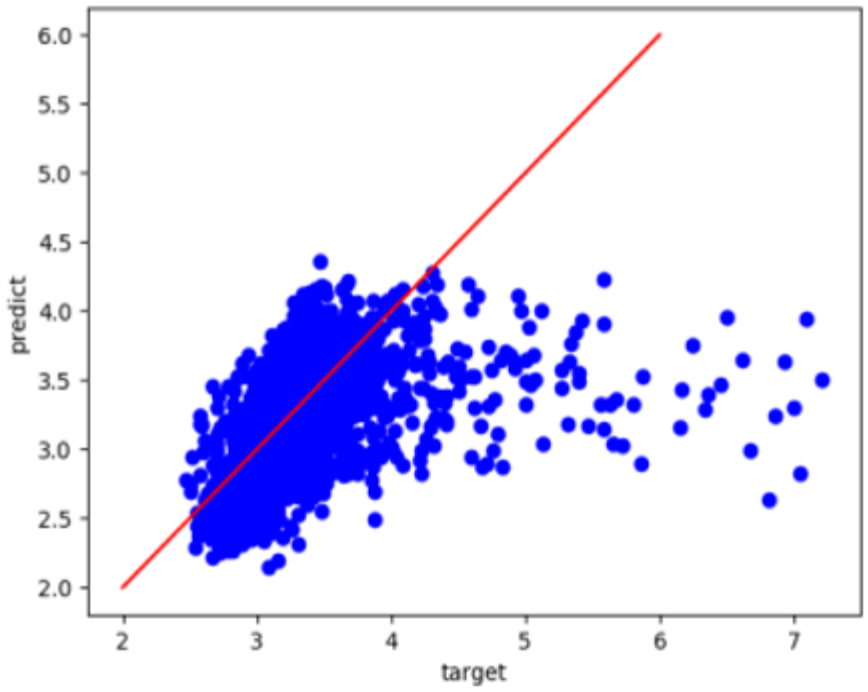
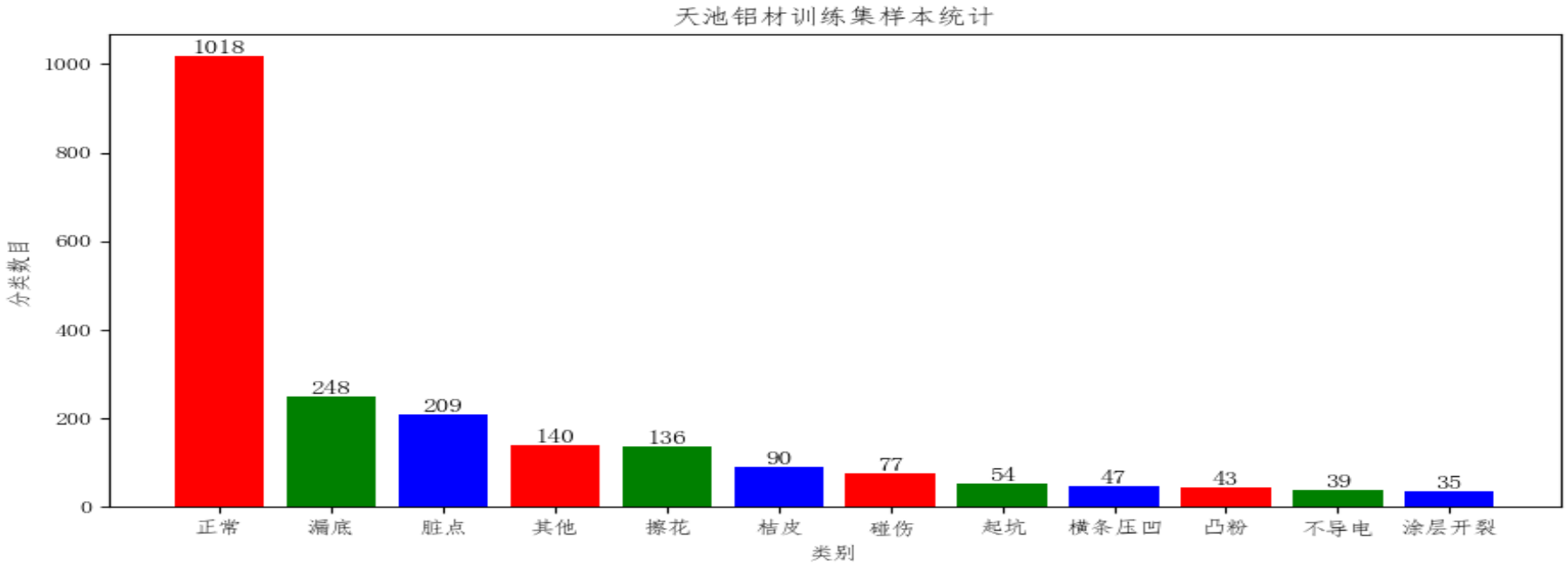
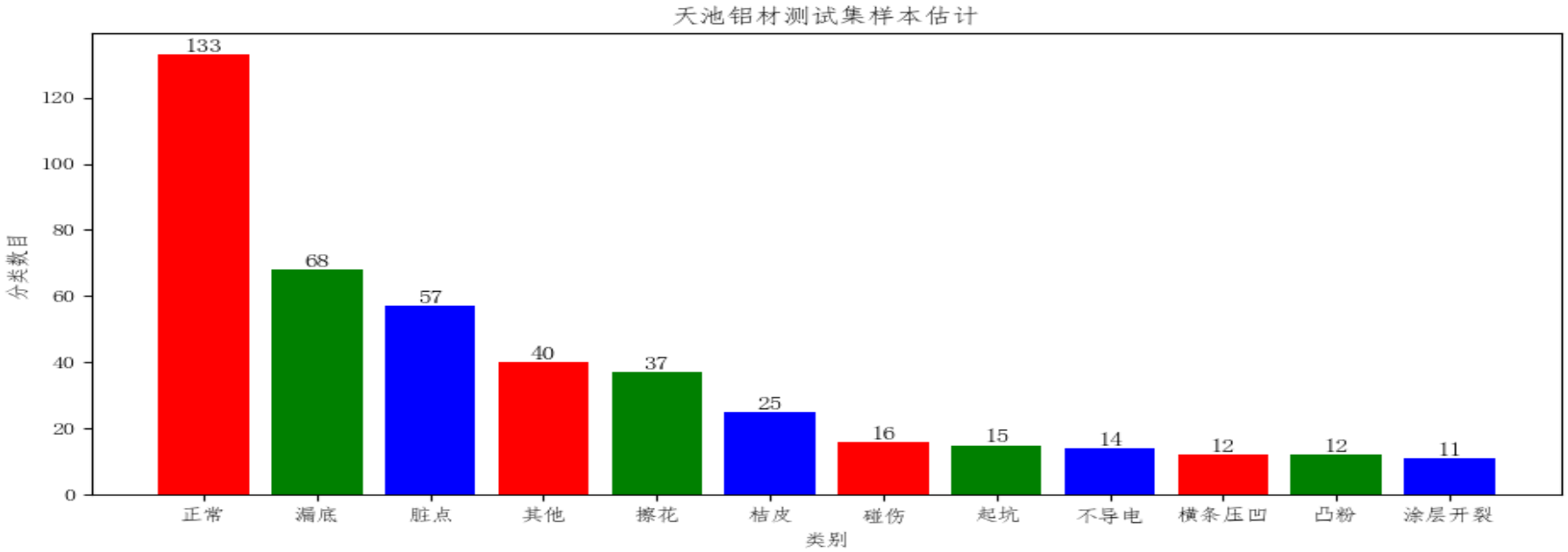


Figure 4. Correlation between ground-truth and predicted difficulty scores. The least squares regression line is almost diagonal suggesting a strong correlation.

http://ruder.io/multi-task/
https://ieeexplore.ieee.org/iel5/69/5173046/05128907.pdf
https://github.com/Vipermdl/Vipermdl.github.io/blob/master/cellar/Domain-Adaptive.pdf


天池铝材训练集样本统计
分类数目
类别
1018
248
209
140
136
90
77
54
47
43
39
35
正常
漏底
脏点
其他
擦花
桔皮
碰伤
起坑
横条压凹
凸粉
不导电
涂层开裂


天池铝材测试集样本估计
分类数目
类别
133
68
57
40
37
25
16
15
14
12
12
11
正常
漏底
脏点
其他
擦花
桔皮
碰伤
起坑
不导电
横条压凹
凸粉
涂层开裂

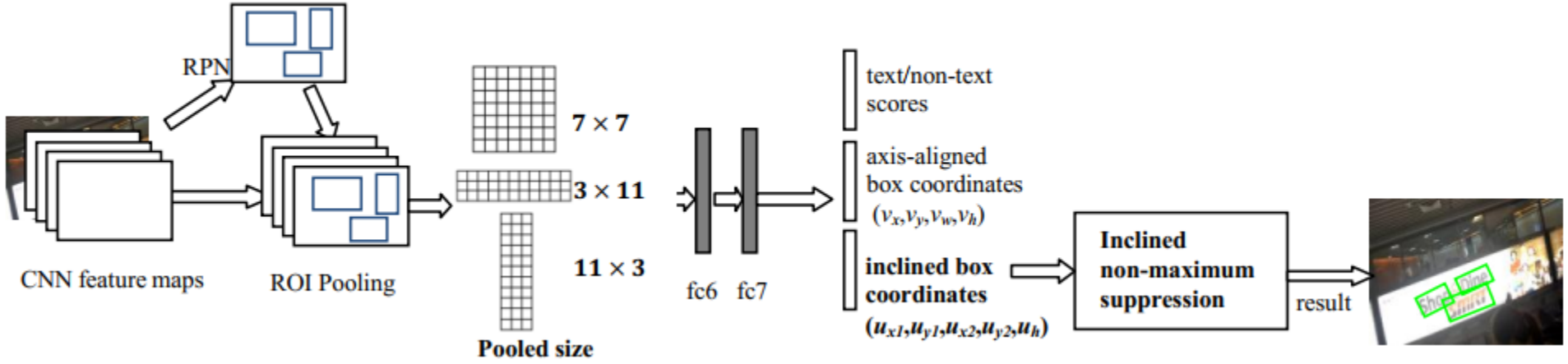
Losses for class imbalance: Since in recent detectors there are a lot of anchors which most of the time cover background, there is a class imbalance between positive and negative anchors. An alternative is Online Hard Example Mining (OHEM). Shrivastava et al. [2016a] performed to select only worst performing examples (so-called hard examples) for calculating gradients. Even if by fixing the ratio between positive and negative instances, generally 1:3, one can partly solve this imbalance. Lin et al. [2017b] proposed a tweak to the cross entropy loss, called focal loss, which took into account all the anchors but penalized easy examples less and hard examples more. Focal loss (Eq. 9) was found to increase the performance by 3.2 mAP points on MS COCO, in comparison to OHEM on a ResNet-50-FPN backbone and 600 pixel image scale.

$$FL(p, y) = \begin{cases} -\alpha_t (1-p)^\gamma log(p) & \text{if } y = 1 \\ -\alpha_t p^\gamma log(1-p) & \text{otherwise} \end{cases} \tag{9}$$

One can also adopt simpler strategies like rebalancing the cross-entropy by putting more weights on the minority class [Ogier Du Terrail and Jurie, 2017].

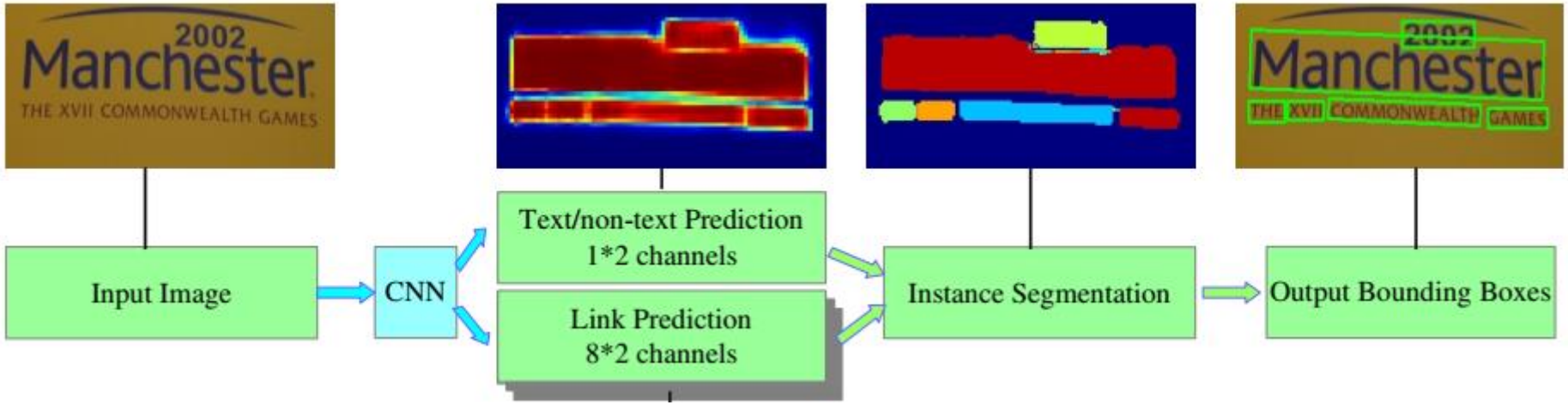
https://arxiv.org/abs/1809.03193

R2CNN



https://arxiv.org/abs/1706.09579
https://arxiv.org/abs/1807.06521
https://github.com/yangxue0827/R2CNN_FPN_Tensorflow

PixelLink



https://github.com/junfu1115/DANet
https://github.com/PkuRainBow/OCNet
https://www.aaai.org/ocs/index.php/AAAI/AAAI18/paper/download/16469/16260

THANK YOU